

Mind the GAP: Geometry Aware Passthrough Mitigates Cybersickness

Trishia El Chemaly
Stanford University
Stanford, California, USA
Google
Mountain View, California, USA
tchemaly@stanford.edu

Mohit Goyal
Google
Mountain View, California, USA
mohitgl@google.com

Tinglin Duan
Google
Mountain View, California, USA
tduan@google.com

Vrushank Phadnis
Google
Mountain View, California, USA
vrushank@google.com

Sakar Khattar
Google
Mountain View, California, USA
sakark@google.com

Bjorn Vlaskamp
Google
Seattle, Washington, USA
bjornvlaskamp@google.com

Achin Kulshrestha
Google
Toronto, Ontario, Canada
kulac@google.com

Eric Lee Turner
Google
Cambridge, Massachusetts, USA
elturner@google.com

Aveek Purohit
Google
Mountain View, California, USA
aveek@google.com

Gregory Neiswander
Google
Mountain View, California, USA
neiswander@google.com

Konstantine Tsotsos
Google
Toronto, Ontario, Canada
ktsotsos@google.com

Abstract

Virtual Reality headsets isolate users from the real-world by restricting their perception to the virtual-world. Video See-Through (VST) headsets address this by utilizing world-facing cameras to create Augmented Reality experiences. However, directly displaying camera feeds can cause visual discomfort and cybersickness due to the inaccurate perception of scale and exaggerated motion parallax. This paper presents initial findings on the potential of geometry aware passthrough systems to mitigate cybersickness through enhanced depth perception. We introduce a promising protocol for quantitatively measuring cybersickness experienced by users in VST headsets. Using this protocol, we conduct a user study to compare direct passthrough and geometry aware passthrough systems. To the best of our knowledge, our study is the first one to reveal reduced nausea, disorientation, and total scores of cybersickness with geometry aware passthrough. It also uncovers several potential avenues to further mitigate visually-induced discomfort.

This research was conducted during Trishia El Chemaly's time at Google as a Student Researcher.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720163>

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; User studies; Usability testing.

Keywords

Video see-through, Cybersickness, Augmented Reality headsets, Motion sickness, View synthesis

ACM Reference Format:

Trishia El Chemaly, Mohit Goyal, Tinglin Duan, Vrushank Phadnis, Sakar Khattar, Bjorn Vlaskamp, Achin Kulshrestha, Eric Lee Turner, Aveek Purohit, Gregory Neiswander, and Konstantine Tsotsos. 2025. Mind the GAP: Geometry Aware Passthrough Mitigates Cybersickness. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3706599.3720163>

1 Introduction

Virtual Reality (VR) head-mounted displays (HMDs) utilize high resolution near-eye displays and multi-sensory input systems to fully immerse the user in a virtual environment.

The high level of immersion isolates the user from their physical surroundings. To allow seamless integration between the real and virtual worlds, Augmented Reality (AR) HMDs overlay digital content onto the reprojected physical world surrounding the user. As a result, in recent years, Video See-Through (VST) or passthrough AR devices such as Apple Vision Pro¹ and Meta Quest 3² have become increasingly popular. VST HMDs utilize camera-captured imagery

¹<https://www.apple.com/apple-vision-pro>

²<https://www.meta.com/quest>

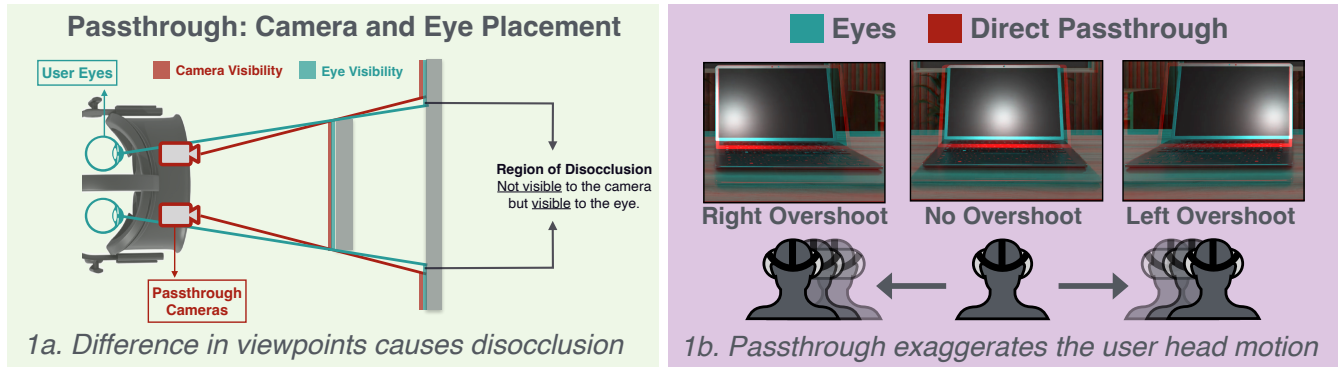


Figure 1: Video see-through headsets employ high resolution cameras to display the user’s physical environment. Due to inherent hardware limitations, the camera’s perspective deviates from the user’s natural viewpoint. Hence, directly displaying camera feeds (direct passthrough) can result in visual artifacts such as disocclusion (1a), inaccurate perception of object positions, and exaggerated motion parallax (1b).

offering users the ability to transition between AR and VR modes to deliver immersive experiences across the full reality-virtuality continuum [30].

As the adoption of VST HMDs as *spatial computers* is increasing for educational,³ gaming[28], and productivity applications, there is a need for research focused on user discomfort and cybersickness associated with VST technology. While motion sickness in VR [5, 7, 26] has been extensively studied in the last few decades, there is limited work dedicated to enhancing comfort and safety with the use of AR devices. Insights from VR research are informative, but the unique nature of VST where users can see and interact with the physical world suggests a need for dedicated investigation to guide the design of VST HMDs that prioritize user comfort.

Due to hardware and physical constraints, cameras are generally positioned on the outer surface of a VST headset, away from user’s eye positions (Figure 1a). *Direct passthrough* (DP), i.e. delivering the raw camera feed to the display, exaggerates distances and consequently the motion of objects (Figure 1b). According to the sensory conflict theory, this mismatch between visual and inertial cues can potentially cause discomfort [31]. To mitigate this mismatch, depth information is utilized to reproject camera image feeds into the eye views [9, 38]. We introduce the term *Geometry Aware Passthrough* (GAP) to describe these depth-based reprojection algorithms. While previous work assumes that GAP reduces discomfort compared to DP, to the best of our knowledge, no empirical studies have directly investigated this assumption. In addition, depth discontinuities and disocclusions can lead to warping artifacts in GAP algorithms, which could further contribute to the overall cybersickness experienced by users.

In our work, we introduce a comprehensive protocol focused on key VST use cases to assess visually-induced discomfort and cybersickness in VST HMDs. We apply this protocol to compare GAP to DP and present our initial findings. The study encompasses three activities in VST: (i) typing on physical keyboards, (ii) navigation around obstacles, and (iii) near-field object interaction. Through the proposed study, our initial results indicate that GAP reduces

nausea, disorientation, and total scores of cybersickness compared to DP. We believe this work lays the foundation for future research focused on designing VST systems with enhanced user comfort.

2 Related Work

Cybersickness and Comfort in VST. Literature on VST discomfort builds on seminal work on simulator sickness, which explores psycho-physical causes like sensory conflict and postural instability [33]. With the rise of VR HMDs, studies have contextualized these findings in VR applications [29]. LaViola et al. [26] studied the visual-vestibular mismatch in VR environments. Blum et al. investigated out-of-focus blur in VR and found it less problematic, noting lower levels of double vision and higher tolerance for blur [5]. While VR sickness research is extensive, VST-specific studies remain limited, focusing on mitigating discomfort through visual displacement conditions [23, 27], fisheye lenses to expand peripheral vision [32], and latency reduction methods [15]. User studies and geometric models demonstrate that displacement of viewer eyes from the intended center of projection contribute to distortions and misperceptions that may induce cybersickness [16, 18, 34, 37]. Early studies have shown that visual displacement, a key characteristic of direct passthrough, can induce significant perceptual costs and adaptation challenges due to the introduced intersensory conflict [4]. More recently, Li et al [29] investigated mixed-reality tunneling methods for mitigating sickness. Our work proposes a comprehensive evaluation framework focusing on user evaluations to address hardware and software aspects of VST.

Reprojection in VST HMDs. For VST HMDs, the passthrough cameras are placed in front of the user’s eyes, typically a few centimeters away. The scene as viewed by the user’s eyes needs to be reconstructed and displayed back to the user through VST displays. This process is generally referred to as *reprojection*, where the camera images are reprojected to the user’s eyes. Past work on performing on-device reprojection for passthrough either utilizes classical depth estimation to synthesize eye-views [9] or relies on dedicated GPUs to perform real-time accurate reprojection [6, 8, 38]. Light field passthrough cameras offer an alternative to traditional

³<https://giantlazer.com>

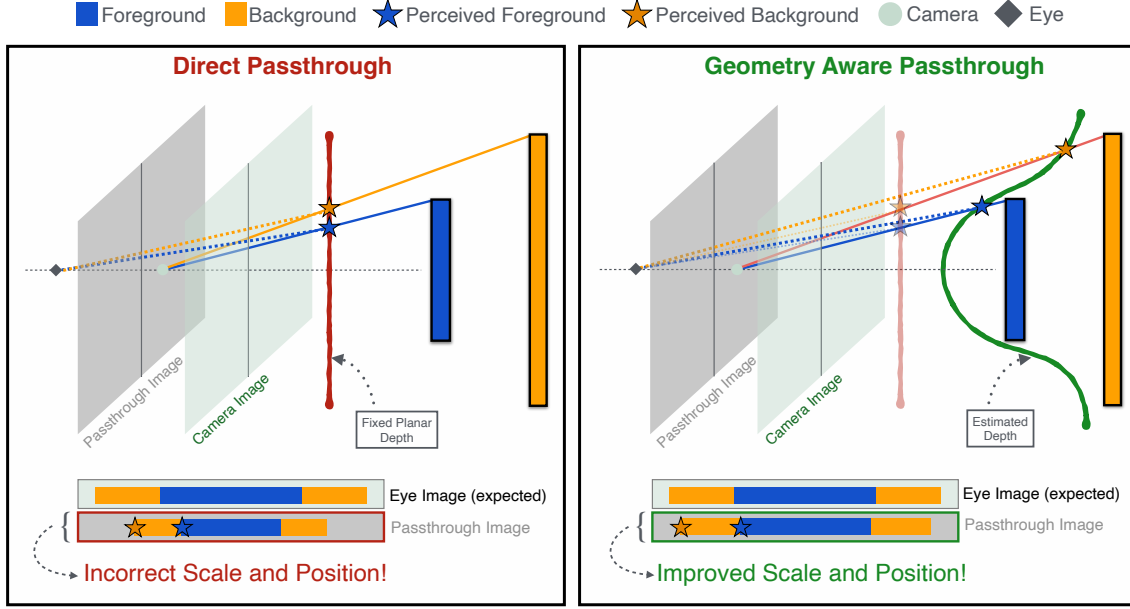


Figure 2: 2D illustration comparing reprojection between direct and geometry aware passthrough. The foreground and background are reprojected using different geometries. In direct passthrough, each 3D point is assumed to belong to a single plane situated at a certain distance. This results in reprojecting objects at incorrect pixel locations and scale (left). In comparison, geometry aware passthrough uses a geometry estimate which improves the perceived location and scale (right).

reprojection methods by directly capturing the perspective to potentially reduce warping artifacts and latency [25]. In our work, we follow Chaurasia et al. [9] and implement our GAP pipeline using a low powered, on-device depth estimation algorithm. While many of these studies discuss several technical aspects that are important for reprojection in headsets, their impact and the significance of reprojection itself on cybersickness remain underinvestigated.

3 Passthrough Modes

In this section, we describe the technical differences between the two passthrough approaches (i) Direct Passthrough (DP) and (ii) Geometry Aware Passthrough (GAP) that we evaluate in Section 4. **Direct Passthrough.** DP directly displays unprocessed video feeds from external cameras, forgoing any form of view correction. However, this inherent simplicity introduces a critical limitation: visual discrepancies in object positions and scales (Figure 2). The disparity between the physical camera locations and the user’s eyes manifests as an exaggerated perception of motion parallax, particularly pronounced when observing objects in close proximity. Theoretically, DP can be interpreted as a simplified form of planar reprojection, where the scene is assumed to be geometrically planar (Figure 2), with the plane being fronto-parallel and situated infinitely far from the cameras. In our practical implementation, we set the projection plane 2 meters from the cameras since most of the objects lie in the 1-2 meters range. This adjustment helps improve the perceived scale of the scene by *better* reflecting the average depth of objects in context of the tasks we chose for cybersickness analysis (Section 4.1). Moreover, it can be implemented using a simple homography

transformation with minimal computational cost, without doing any geometry based correction.

Geometry Aware Passthrough. In contrast to DP, GAP employs novel view synthesis techniques or reprojection to align external camera images with the user’s viewpoint, thereby mitigating the inherent camera-eye disparity. While this enhances visual fidelity, it increases computational demands and can introduce latency. Even with perfect geometric estimations, reprojection may result in holes in the rendered imagery due to depth disocclusion which occurs when objects closer to the camera obscure portions of the scene that would be visible from the user’s perspective. GAP is susceptible to two main artifact types: geometric (due to depth inaccuracies leading to incorrect positions and scales) and warping (from filling holes via stretching at depth discontinuities, Figure 3). Consumer devices, notably Meta Quest, use GAP. Our implementation uses stereo depth estimation for the left camera and generates right camera depth maps using known calibration parameters. This choice of depth estimation method is not fundamental to GAP; it can be replaced with other techniques without compromising the overall methodology. We follow Chaurasia et al. [9] as our approach is similar to their passthrough implementation – *Passthrough+*.

Hardware Details. The studies were conducted using a mixed-reality headset powered by Android XR¹. We strive to keep all factors other than reprojection such as cameras, image processing, and latency constant across experiments. We did not observe a significant difference in latency between DP and GAP, with DP being approximately 1 ms faster than GAP .

¹<https://www.android.com/xr/>

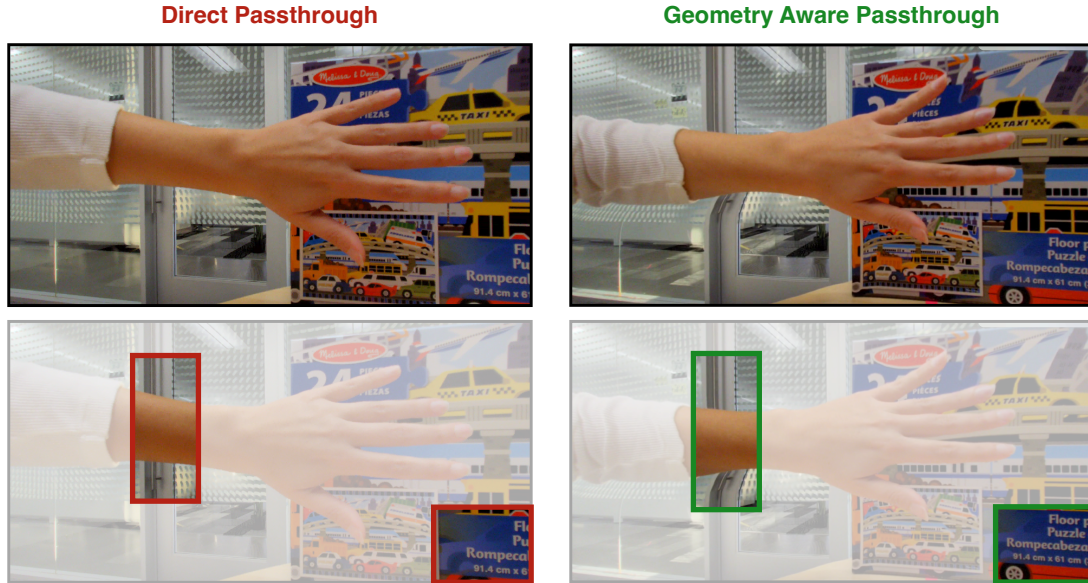


Figure 3: DP versus GAP output images. Two images taken from the HMD at the same pose. DP enlarges objects, making the scene look closer to the user. The scale difference can be noticed when comparing the text on the bottom right of both images. While GAP improves the scale, it results in warping artifacts which can be observed on the edge of the door behind the hand.

4 User Study

We introduce a protocol specifically tailored to quantify cybersickness in the context of VST HMD use cases. This section elucidates the design of the protocol, encompassing task selection and quantification methodologies. A within-subject design is employed to gather both quantitative and qualitative subjective feedback across three distinct visual conditions: Natural Vision (NV), where participants complete the tasks without a headset; Direct Passthrough (DP); and Geometry Aware Passthrough (GAP).

Participants. We recruited 25 participants who had normal or corrected-to-normal vision. The study included a diverse range of ages, genders, levels of VR usage, and job profiles. Table 1 in Appendix A.1 gives a full overview of the demographics. Recruitment was done in accordance with the ethics board of our institution. We excluded data from one participant from our analysis since they spent an abnormally long time to complete the protocol.

4.1 Tasks

We devised our protocol focusing *exclusively* on passthrough-based real-world interactions and ensured no virtual elements were visible [11]. Figure 4 shows pictures of the lab setup. The tasks were inspired from fundamental real-world AR applications. They emphasized on the user head motion while necessitating inspection and spatial awareness of the physical world. Our design ensured a cumulative duration to be roughly 15 minutes for all the tasks.

Typing. Typing was chosen to simulate applications in productivity such as using VST HMDs like the Apple Vision Pro as extensions of traditional work setups where users interact with physical keyboards and screens. It engages both visual and motor components

and represents fine interactions requiring frequent gaze shifts between the keyboard and the laptop screen, a common trigger for visual discomfort. A physical Dvorak keyboard [12] was used to minimize reliance on muscle memory and encourage frequent visual engagement with the keys. Participants typed for 6 minutes as quickly and accurately as possible, with the typing text randomized for each session. The placement of the laptop and keyboard was standardized, and screen brightness was kept consistent to ensure uniformity across participants.

Navigation. The realistic and holistic use of a VST HMD involves navigating physical spaces, avoiding obstacles, and interacting with real-world objects [3, 10, 13]. Previous research has explored the impact of spatial navigation, such as waypoint-following, on sickness in VR [2]. We designed a navigation task where participants collected and dropped off 10 numbered cones, one at a time, into a designated drop zone. To emphasize geometry perception, the task included narrow passages and required multi-directional movements, including turning to locate cones and interacting with objects at varying heights. Participants were given 2-3 minutes to familiarize themselves with the cone locations before beginning the trial. They were instructed to move naturally but carefully to avoid colliding with objects in the room.

Interaction. Li et al. [29] introduced a Tangram puzzle task to simulate assembly tasks requiring both motor and cognitive skills. Inspired by their approach, we adapted this task by using 2 x 3 feet jigsaw puzzles consisting of 24 pieces. This larger size was selected to accommodate head motion which is often associated with motion sickness. To maintain a consistent level of difficulty across conditions, we utilized three different jigsaw puzzles from

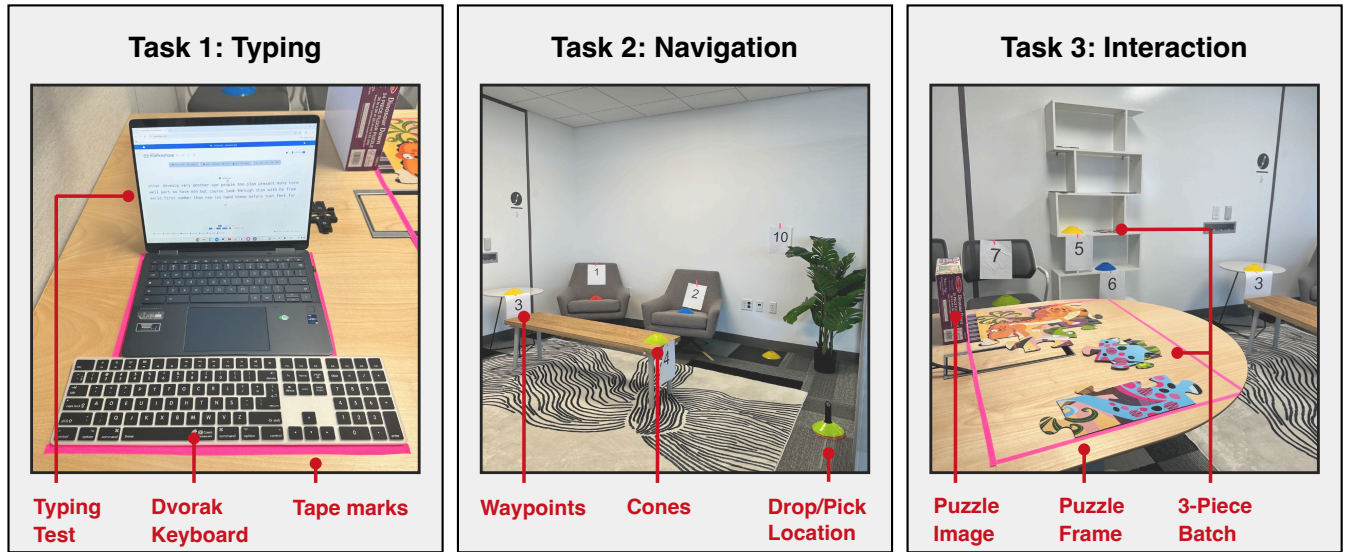


Figure 4: User Study Setup. Pictures of the lab setup for the tasks completed by participants while wearing the VST HMD.

the same series by the same manufacturer. To standardize the puzzle completion strategy, we divided the puzzle pieces into 8 batches placed on the same shelves for all trials. Participants were restricted to retrieving and working on only one batch at a time within a rectangular frame marked on the table. This setup allowed us to incorporate locomotion into the task as participants moved between batches. The puzzles were assigned to the conditions randomly but were not repeated.

4.2 Measures

Quantitative Measures. The Simulator Sickness Questionnaire (SSQ), widely regarded as the standard for measuring VR sickness [19, 20, 22, 24, 36], was used to capture 16 symptoms across four subscales: Nausea, Disorientation, Oculomotor, and Total Severity. Participants rated symptom severity on a 4-point Likert scale (0 = none, 1 = slight, 2 = moderate, 3 = severe). SSQ scores were collected right before and after each condition, allowing us to isolate the SSQ per condition by evaluating the difference between post and pre condition scores, as evidenced in prior literature [29]. Discomfort ratings were also collected immediately after each task while participants were still wearing the headset. Discomfort was defined as any sensation that might prompt participants to leave the setup [14], including SSQ symptoms. Participants responded to the question: “On a scale of 0 to 10, where 0 represents how you felt before starting and 10 means you want to stop, how do you feel now?” This approach, adapted from Fernandes and Feiner [14], has been previously employed to facilitate real-time monitoring of discomfort [1, 15]. The average discomfort score was calculated using all the collected scores.

Qualitative Feedback. At the end of each trial, participants answered open-ended survey questions about their experiences. Interviews began with discussing preferences between VST conditions, followed by detailed reasoning behind those preferences. NV was

not included in the preference question to focus on comparing passthrough conditions.

4.3 Procedure

Prior to the trial, participants signed a consent form and completed a demographics questionnaire. Their IPD was measured with an optical digital pupilometer, and the HMD was adjusted accordingly. The experimenter provided task instructions and demonstrated the procedures. Participants were randomly assigned to one of six condition orders, determined using a balanced Latin square design. Each condition involved completing the typing, navigation, and interaction tasks in a fixed sequence to focus on overall cybersickness rather than task-specific effects.

Participants wore the headset continuously for each condition until all tasks were completed. The experimenter recorded discomfort scores and key observations throughout. Participants filled out the SSQ before and after each condition. To mitigate carryover effects, participants were required to take a 15-minute break between conditions, with access to water and a space with windows. The entire session, including two breaks, lasted approximately 90 minutes.

5 Results and Discussion

Our work focuses on the impact of reprojection in passthrough on cybersickness and discomfort. Below, we discuss key findings from our user study, highlight implications for enhancing VST systems, and outline potential future directions.

GAP Mitigates Cybersickness. We calculate the SSQ sub-scores using the method by Kennedy et al. [21] and use the post and pre-condition score differences as the dependent variable. The statistical analysis is summarized in Table 5.

The Shapiro-Wilk test indicated that the SSQ scores were not normally distributed ($p < 0.001$). Consequently, Friedman test was

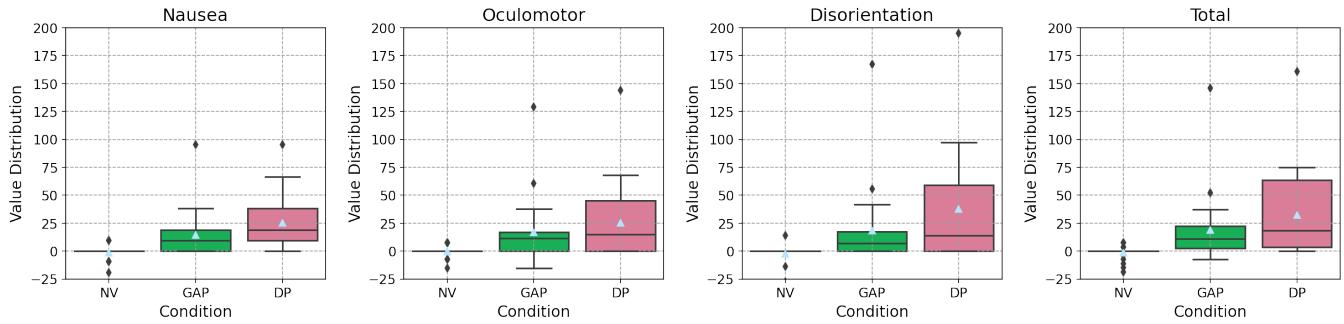


Figure 5: SSQ. Box plots of Nausea (N), Oculomotor (O), Disorientation (D), and Total subscores of simulator sickness comparing all conditions. NV led to significantly lower N, O, D, and total sickness scores than both DP and GAP ($p < 0.05$). GAP led to significantly lower N ($p = 0.016$), D ($p = 0.029$), and total ($p = 0.011$) sickness scores than DP. Black diamonds represent outliers, blue diamonds represent mean values, and the line inside each box represents the median.

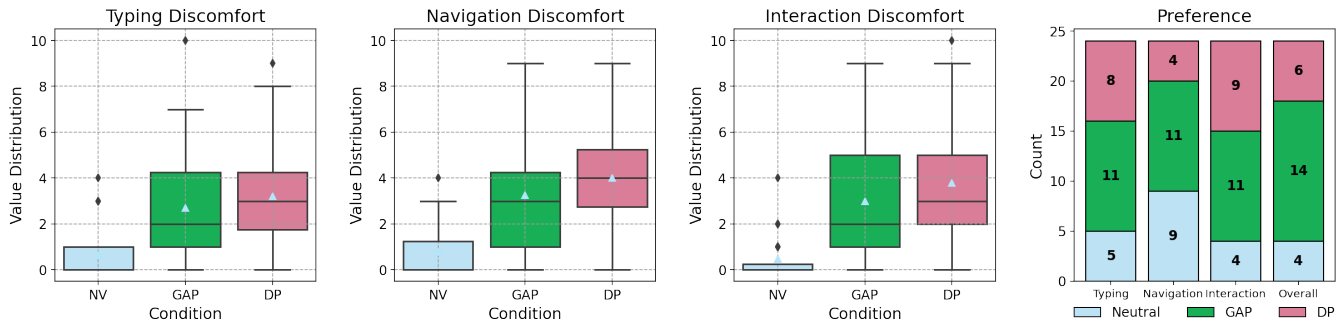


Figure 6: Subjective Discomfort. Box plots of discomfort scores for all conditions and preference for passthrough conditions across the typing, navigation, and interaction tasks. NV led to significantly lower discomfort scores than both DP and GAP ($p < 0.05$) across all tasks. Compared to DP, GAP led to significantly lower discomfort scores across all tasks, i.e., typing ($p = 0.046$), navigation ($p = 0.041$), interaction ($p = 0.022$). Black diamonds represent outliers, blue diamonds represent mean values, and the line inside each box represents the median.

conducted for each SSQ sub-score, which showed significant differences between conditions for Nausea ($\chi^2 = 31.89$, $p < 0.001$), Oculomotor ($\chi^2 = 25.97$, $p < 0.001$), Disorientation ($\chi^2 = 23.89$, $p < 0.001$), and Total Severity ($\chi^2 = 33.82$, $p < 0.001$). To identify specific condition differences, we performed pairwise Wilcoxon signed-rank tests with Holm-Bonferroni adjustment. NV resulted in significantly lower Nausea, Oculomotor, Disorientation, and Total Severity scores compared to both DP and GAP ($p < 0.05$). GAP led to significantly lower Nausea ($p = 0.016$), Disorientation ($p = 0.029$), and Total Severity ($p = 0.011$) scores than DP. Although GAP had lower mean Oculomotor scores compared to DP, the difference was not statistically significant. We also identified that GAP exposed participants to low levels of cybersickness while DP exposed participants to moderate levels of cybersickness [17, 35]. Figure 5 displays the distribution of simulator sickness scores for all three conditions across sub-scores.

Subjective discomfort scores also showed non-normal distribution using the Shapiro-Wilk test ($p < 0.001$). Friedman tests revealed significant differences in discomfort scores between conditions for typing ($\chi^2 = 26.60$, $p < 0.001$), navigation ($\chi^2 = 28.00$, $p < 0.001$),

interaction ($\chi^2 = 33.34$, $p < 0.001$), and average discomfort ($\chi^2 = 33.23$, $p < 0.001$). Pairwise Wilcoxon signed-rank tests with Holm-Bonferroni adjustment demonstrated that NV led to significantly lower discomfort scores across all tasks compared to both DP and GAP ($p < 0.05$). NV also resulted in significantly lower average discomfort scores compared to both DP and GAP ($p < 0.05$). GAP led to significantly lower discomfort scores across all tasks compared to DP, including typing ($p = 0.046$), navigation ($p = 0.041$), and interaction ($p = 0.022$), as well as significantly lower average discomfort scores ($p = 0.016$). Figure 6 illustrates the distribution of discomfort scores for all three conditions across tasks, with navigation showing the highest mean discomfort, followed by interaction.

Sensory Conflict in DP Causes Discomfort. Several participants reported a mismatch between vision and motion with DP, aligning with the sensory conflict theory and supporting our hypothesis that DP exaggerates motion effects. P17 described their experience: "What I see does not match what is happening...Do you trust what you see or how you move? These don't overlap, and I don't know which one to trust." P12 also noted, "[DP] is worse when there is head motion,

especially when completing the puzzle, because everything feels like it is moving."

Geometry Enhances Spatial Awareness. The researcher observed that participants moved closer to cones during the navigation task with DP due to inaccurate depth cues. Participants confirmed impaired spatial awareness and depth perception with DP compared to GAP. For example, P17 said, *"Depth perception was worse with [DP], and I tested that with the cone experiment. With [GAP], I could toss the cone in and not miss but with [DP] I had to be more careful about where to put the cone."* Additionally, unstable gait and more frequent collisions with furniture were noted in DP: *"I almost knocked things over. I had to be more cognizant of my steps around the furniture. With [GAP], I did not have that problem."*

Task-Specific Observations. Participants were asked to indicate their preference between DP and GAP or to indicate no preference. NV was not included in the preference question to focus on comparing passthrough conditions. Based on overall experience, GAP was generally preferred over DP (Figure 6). Participants preferred GAP for typing (GAP: 46% vs. DP: 33%), navigation (GAP: 46% vs. DP: 17%), and interaction (GAP: 46% vs. DP: 38%). Varying levels of cybersickness were experienced depending on the task. Specifically, many preferred GAP for navigation due to a noticeable mismatch between visual and inertial cues, which was evident in their unstable gait and impaired spatial awareness. In contrast, some participants were more affected by head motion as they assembled the puzzle. While most participants preferred GAP for all tasks, some favored DP for typing due to warping issues on the keyboard: *"I prefer GAP for movements but DP for typing because of the distortions on the keys"* (P1).

Image Quality Tradeoffs. While overall geometry is more accurate with GAP, significant distortions at occlusion boundaries (Figure 3) impact user preference. As highlighted in the preference panel in Figure 6, nearly half of the participants who preferred one passthrough mode over the other chose DP in the interaction condition. Among the 25 participants, 18 explicitly reported reasons for disliking a passthrough mode, with 10 citing warping artifacts as a key issue. This finding underscores the need to measure and minimize these distortions for improved passthrough experiences. Future work should further investigate the balance between geometric accuracy and artifact reduction to optimize usability.

Avenues for Future Research. Beyond reprojection, a wide range of factors must be addressed to enhance comfort in VST HMDs. These include latency and temporal consistency of scene updates, field of view, physical fit of the device, and image quality parameters such as noise, sharpness, color accuracy, and camera optics. This is reflected in the qualitative feedback where latency, frame drops, overexposed images, and blurry vision are recurrently mentioned. Notably, blurry vision emerged as the most commonly reported issue with a higher incidence in DP. Minimizing discrepancies between natural vision and VST HMDs is crucial for ensuring a comfortable experience and facilitating adoption. This suggests the need for more research to understand the impact of these factors on mitigating user discomfort and cybersickness.

6 Limitations

One limitation of our study is the within-subjects design with short washout periods between conditions, which may have introduced

carry-over effects. This design choice was made to directly capture user preferences across conditions while balancing practical constraints related to participant recruitment and retention. We implemented counterbalancing and provided a 15-minute break between conditions. Although participants were informed that they could extend the break if needed, none of them did. We also analyzed SSQ score differences to isolate cybersickness impact. However, some participants showed an improvement in symptoms throughout the course of the NV condition. This suggests that the break period may have been insufficient to reset cybersickness symptoms. Future studies should consider longer washout periods or between-subjects designs to further minimize residual effects. A mixed-effects linear model could further provide a more robust analysis of the impact of session ordering on cybersickness.

Given the well-known individual differences in susceptibility to cybersickness, the sample size of 25 participants may not fully capture the variability across a larger population. Future research should consider larger sample sizes to enhance the generalizability of the findings. Furthermore, incorporating a Visually Induced Motion Sickness (VIMS) susceptibility survey could help account for individual differences in cybersickness susceptibility.

In addition, the task durations in this study were short, particularly for navigation and puzzle-solving tasks, which may have limited the extent of cybersickness symptoms observed. Future studies should explore longer task durations to provide a more comprehensive assessment of cybersickness effects.

The repeated administration of the SSQ within a within-subjects design can potentially inflate reported symptom severity. While this study followed a standardized protocol, future research should aim to minimize the number of SSQ repetitions and adhere to best practices for reducing response inflation effects.

While GAP demonstrated lower cybersickness symptoms compared to DP, the underlying mechanisms require further investigation. Therefore, the relationship between depth perception and cybersickness is not yet fully understood. Factors such as sensory conflict theory, postural instability theory, and the rest frames hypothesis may play a role. Future research should explore these theories further to better understand why GAP reduces discomfort despite introducing warping and disocclusion artifacts.

7 Conclusion

In this work, we investigated the impact of reprojection algorithms on user discomfort and cybersickness in VST HMDs. We presented a promising protocol aimed at evaluating visually-induced discomfort and cybersickness in VST HMDs through key use cases. Our initial findings suggest that GAP reduces nausea, disorientation, and total scores of cybersickness as well as subjective discomfort scores as compared to DP. Notably, our findings revealed that cybersickness in VST systems is a complex phenomenon, distinct from both VR cybersickness and traditional motion sickness. Although the levels of cybersickness observed were generally low to moderate, these symptoms can influence user acceptance and overall experience. We believe this work provides an initial step toward refining these systems and enhancing user comfort in VST technologies.

Acknowledgments

To Abhishek Kar for his guidance during ideation of this work.

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A APPENDIX

A.1 PARTICIPANT DEMOGRAPHICS

Table 1: Participant demographics for the user study, showing diversity across gender, age, normal versus corrected vision, and VR usage.

Variable	Categories	#Participants
Gender	Men	14
	Women	11
Age Group	18-24	1
	25-34	12
	35-44	9
	45-54	3
Vision	Normal	19
	Contact Lenses	6
VR Usage	Never	1
	Once	12
	Once a week	3
	Once a month	8
	At least once a day	1

A.2 RESULTS

A.2.1 Simulator Sickness Scores. Participants rated 16 SSQ symptoms (Table 2) on a 4-point Likert scale (0 = none, 1 = slight, 2 = moderate, 3 = severe) before and after each condition. SSQ sub-scores were computed following the method of Kennedy et al. [21]. The equations for SSQ sub-score calculations are provided below Table 2. We used the difference between post- and pre-condition sub-scores to construct Figure 5 and Table 3.

As shown in Table 3, the mean SSQ sub-scores for NV were negative, indicating that some participants reported a reduction in symptom severity after the NV condition. This suggests that the washout period may have been insufficient, and that the NV condition contributed to symptom alleviation. However, some participants reported symptom improvement even when NV was the first condition in their trial. This variability could also be attributed to the subjective nature of SSQ assessments. In Figure 5, the black diamonds represent outliers in the data. We include them in the visualization to ensure a comprehensive representation of the data distribution.

Table 2: Simulator Sickness Questionnaire (SSQ) Symptoms and Weights

SSQ Symptoms		Weight		
		N	O	D
1	General Discomfort	1	1	
2	Fatigue		1	
3	Headache		1	
4	Eyestrain		1	
5	Difficulty Focusing		1	1
6	Increased Salivation	1		
7	Sweating	1		
8	Nausea	1		1
9	Difficulty Concentrating	1	1	
10	Fullness of Head			1
11	Blurred Vision		1	1
12	Dizzy (Eyes Open)			1
13	Dizzy (Eyes Closed)			1
14	Vertigo			1
15	Stomach Awareness	1		
16	Burping	1		
Total		[1]	[2]	[3]

The pre-condition and post-condition **nausea (N)**, **oculomotor (O)**, **disorientation (D)**, and **total score (TS)** sub-scores were computed as follows:

$$N = [1] \times 9.54 \quad (1)$$

$$O = [2] \times 7.58 \quad (2)$$

$$D = [3] \times 13.92 \quad (3)$$

$$TS = ([1] + [2] + [3]) \times 3.74 \quad (4)$$

The **reported sub-scores** used to construct Figure 5 and Table 3 are computed as follows:

$$\text{Reported sub-score} = \text{Post-condition sub-score} - \text{Pre-condition sub-score} \quad (5)$$

Table 3: Mean, standard deviation, median, and interquartile range of simulator sickness for all conditions

Condition	Nausea (N)				Oculomotor (O)				Disorientation (D)				Total Score (TS)			
	Mean	SD	Median	IQR	Mean	SD	Median	IQR	Mean	SD	Median	IQR	Mean	SD	Median	IQR
NV	-1.19	5.72	0	0	-0.63	4.85	0	0	-2.32	8.68	0	0	-1.40	5.99	0	0
DP	25.44	22.82	19.08	28.62	25.27	33.09	15.16	45.48	37.70	47.96	13.92	59.16	32.57	36.79	18.70	59.84
GAP	14.71	20.05	9.54	19.08	17.06	27.79	11.37	17.05	18.56	34.72	6.96	17.40	19.17	29.97	11.22	19.64

Table 4: Mean and standard deviation of subjective discomfort across tasks for all conditions

	Typing		Navigation		Interaction		Average	
Condition	Mean	SD	Mean	SD	Mean	SD	Mean	SD
NV	0.63	1.11	0.79	1.12	0.46	0.96	0.63	0.99
DP	3.21	2.47	4.00	2.57	3.79	2.63	3.67	2.38
GAP	2.71	2.57	3.25	2.73	3.00	2.65	2.99	2.52

Table 5: The results of statistical analysis for the SSQ and subjective discomfort scores (* $p < 0.001$; ** $p < 0.05$)

Dependent Variables		Friedman	Wilcoxon signed-rank
SSQ	Nausea	$\chi^2 = 31.89^*$	DP > NV**, GAP > NV**, DP > GAP **
	Oculomotor	$\chi^2 = 25.97^*$	DP > NV**, GAP > NV**
	Disorientation	$\chi^2 = 23.89^*$	DP > NV**, GAP > NV**, DP > GAP **
	Total	$\chi^2 = 33.82^*$	DP > NV**, GAP > NV**, DP > GAP **
Discomfort	Typing	$\chi^2 = 26.60^*$	DP > NV**, GAP > NV**, DP > GAP **
	Navigation	$\chi^2 = 28.00^*$	DP > NV**, GAP > NV**, DP > GAP **
	Interaction	$\chi^2 = 33.34^*$	DP > NV**, GAP > NV**, DP > GAP **
	Average	$\chi^2 = 33.23^*$	DP > NV**, GAP > NV**, DP > GAP **